8.3 Harvesting a Population

WORKSHEET

The question

STEP II 1995, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q8

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Before solving anything, sketch the growth rate $(2K - x)x$ as a function of x. Where is it zero? Where positive? What does that tell you about how x will behave?

STAGE 1 Solving the logistic equation

Covers: the population without removal

Ask yourself Separate the variables and split $\dfrac{1}{(2K - x)x}$ into partial fractions.

Ask yourself Use $x = K$ at $t = 0$ to find the constant. Then rearrange for x and see what happens as $t \to \infty$.

STAGE 2 **Harvesting**

Covers: with removal at rate L

Ask yourself Removing bacteria at rate L subtracts L from the rate of growth. Complete the square in $(2K - x)x - L$, and write $K^2 - L = \alpha^2$.

Pause and try Substitute $y = x - K + \alpha$. What equation does y satisfy, and what is $y(0)$? Compare with the first part, with α in place of K.

What's really going on

Tipping points

The harvested equation has two equilibria, $K + \alpha$ and $K - \alpha$. The upper one is *stable*: a population slightly above or below it returns to it. The lower one is *unstable*: a population just below $K - \alpha$ declines to extinction. As the harvest rate L increases, $\alpha = \sqrt{K^2 - L}$ shrinks, and the two equilibria approach each other; at $L = K^2$ they merge, and for $L > K^2$ there is no equilibrium at all, and every population collapses (Figure 8.3.1). This is a *saddle-node bifurcation*, the simplest kind of *tipping point*. Its sting is that nothing dramatic warns of it: as L approaches K^2, the equilibrium population declines only gently, and then suddenly there is no equilibrium. Reducing the harvest afterwards may not help, because the population may already be below the unstable threshold.

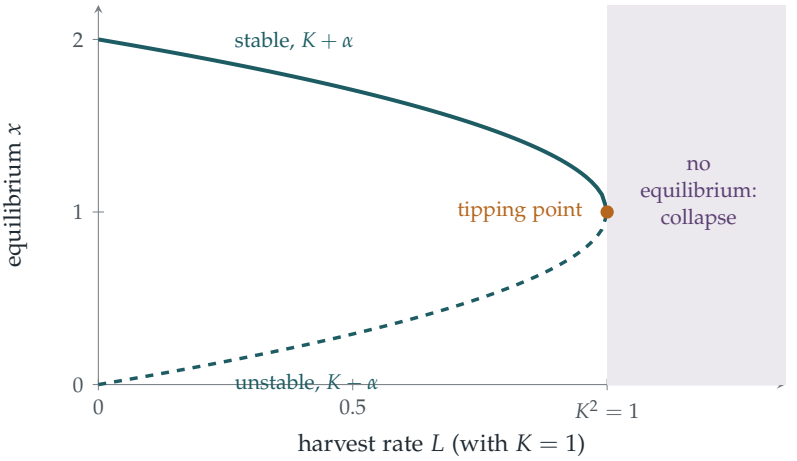


Figure 8.3.1 Equilibrium populations against the harvest rate L, with $K = 1$. The stable equilibrium (solid) and the unstable threshold (dashed) approach each other as L increases, and meet at $L = K^2$. Beyond that, no equilibrium exists and the population collapses whatever its size.

Maximum sustainable yield

What is the largest harvest the population can sustain? It is $L = K^2$, the peak of the growth rate $(2K - x)x$, achieved when the population is held at K, half the carrying capacity. Fisheries science adopted this "maximum sustainable yield" as a management target in the mid-twentieth century. But harvesting exactly at the maximum puts the system exactly at the tipping point, where the slightest overestimate of K, or a bad year, causes collapse. The Grand Banks cod fishery off Newfoundland, one of the richest in the world for five hundred years, collapsed in the early 1990s, and in 1992 the Canadian government imposed a moratorium that put tens of thousands of people out of work. The stocks have still not fully recovered.

Logistic growth elsewhere

The logistic equation was proposed by Pierre-François Verhulst in 1838 to model human populations. The same S-shaped curve describes the spread of an epidemic in its early stages, the adoption of new technologies, autocatalytic chemical reactions and the growth of tumours. Its discrete-time version, the logistic *map*, behaves very differently: it can be chaotic, as we will see in Section 8.7.

Try this next

1 **Starting low.** With harvesting and $K^2 > L$, show that if $x(0) < K - \alpha$ then the population reaches zero in finite time. (*Hint:* what does y do when $y(0) < 0$?)

2 **Over-harvesting.** If $L > K^2$, write $\beta = \sqrt{L - K^2}$. Solve $x' = -(x - K)^2 - \beta^2$ with $x(0) = 2K$ and find when the population reaches zero.

3 **Proportional harvesting.** Suppose instead that a fixed *fraction* of the population is removed: $x' = (2K - x)x - hx$. Find the equilibria and show that the population survives for all $h < 2K$. Why is this policy safer?

4 **Half-time.** In the unharvested case, find the time at which the population reaches $\frac{3}{2}K$.

Further reading

- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). Chapter 3 is about bifurcations, with this harvesting model as an exercise.
- **Nicholas Britton, *Essential Mathematical Biology*** (Springer). Population models for undergraduates.
- **Mark Kurlansky, *Cod: A Biography of the Fish That Changed the World*** (Penguin). The history of the cod fishery and its collapse.

The full guided solution is in Section 8.3:
https://beyondstep.keithmcnulty.org/ch08/harvesting.html